PRODUCTION RUNTIME & DEPLOYMENT SPECIFICATION

KERF 3D STUDIO

Precision Additive Manufacturing · Server Handover Documentation

HOSTING SPEC V2.4

STACK: CodeIgniter 4.7 / PHP 8.2+
DATABASE: MariaDB 10.11+
DATE: October 2026

> ✦ **ZERO-CONFLICT HOST ISOLATION**
> Kerf 3D Studio is designed to be **100% self-contained**. When deployed on a shared VPS with other existing websites, it runs inside lightweight isolated containers or a dedicated virtual host, ensuring **zero port conflicts, zero dependency interference, and zero file pollution**.

01 // SYSTEM ARCHITECTURE & DIRECTORY MAP

The codebase strictly adheres to CodeIgniter 4 framework separation. Only the `public/` folder is exposed to incoming HTTP traffic:

DIRECTORY / FILE	PURPOSE	HOSTING PERMISSION & SECURITY RULE
`app/`	Core App Logic	Controllers, Models, Views. Kept completely outside web access.
`public/` ★	**Web Document Root**	**The ONLY folder exposed to Nginx / Apache / OpenLiteSpeed.**
`public/uploads/cad/`	Temporary CAD Ingestion	Strictly stores incoming STL/OBJ 3D uploads. Must be writable (775).
`writable/`	Runtime Data	Logs, session cache, debug bar. Must have full write permission (775).
`config/credentials/`	Google Drive API Key	Contains `drive-service-account.json` for media synchronization.
`.env`	Environment Secrets	Domain URL, DB credentials, MFS bKash/Nagad accounts. Never committed.
`deploy.sh` / `deploy.ps1`	Hot Reload Scripts	1-Click zero-downtime updates: git pull, migrations, cache clear, reload.

> ★ **ZERO-BANDWIDTH MEDIA STORAGE**
> All heavy product catalog photos are hosted on **Google Drive** and delivered directly to users via Google CDN edge nodes. **Zero storage load and zero bandwidth cost** on the server disk!

02 // DEPLOYMENT METHOD A: DOCKER / VPS (RECOMMENDED)

This is the fastest, cleanest approach. Kerf 3D runs in its own isolated network.

STEP 1 **Clone the Git Repository**

```
git clone https://github.com/redwan2003-bot/KERF_3D_store.git /var/www/kerf3d
cd /var/www/kerf3d
```

STEP 2 **Configure Environment (.env)**

```
cp .env.example .env
nano .env
```

Verify the following production settings inside `.env`:

```
CI_ENVIRONMENT = production
app.baseURL = 'https://kerf3d.store/'   # Replace with actual live domain

# MariaDB credentials (internal to Docker bridge network)
database.default.hostname = db
database.default.database = kerf3d
database.default.username = kerf3d_user
database.default.password = kerf3d_secret_password_here

# MFS Payment Intake
BKASH_ACCOUNT_NUMBER = '01754550297'
NAGAD_ACCOUNT_NUMBER = '01754550297'
```

STEP 3 **Place Google Drive API Key**

Copy the service account credentials provided by the site owner into:

```
/var/www/kerf3d/config/credentials/drive-service-account.json
```

STEP 4 **Launch Docker Containers & Run Migrations**

If host port `8080` is already used by another service, change the mapped port in `docker-compose.yml` (e.g. `"8088:80"`):

```
docker compose up -d
docker exec -w /var/www/vhosts/localhost/html kerf3d_ols php spark migrate --all
```

STEP 5 **Nginx Reverse Proxy & SSL Certbot**

Add this virtual host block to your host Nginx configuration (e.g., `/etc/nginx/sites-available/kerf3d`):

```
server {
    server_name kerf3d.store www.kerf3d.store;

    location / {
        proxy_pass http://127.0.0.1:8080;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        client_max_body_size 128M; # Supports high-poly STL customer uploads
    }
}
```

Issue SSL Certificate:

```
certbot --nginx -d kerf3d.store -d www.kerf3d.store
```

03 // DEPLOYMENT METHOD B: CPANEL / STANDARD HOSTING (NO DOCKER)

- **Document Root:** In cPanel / DirectAdmin, point the domain DocumentRoot specifically to `/path/to/kerf3d/public`. *(Never point directly to the project root directory).*
- **PHP Version:** Select **PHP 8.2** or **PHP 8.3** with extensions: `php-intl`, `php-mbstring`, `php-curl`, `php-mysql`.
- **Permissions:** Grant write permissions: `chmod -R 775 writable public/uploads`.
- **Database:** Create MySQL DB in cPanel, enter details in `.env`, then run `php spark migrate --all` via SSH or Terminal.

04 // AUTOMATED MEDIA SYNC (CRON JOB)

To automatically sync newly uploaded photos from Google Drive into the live catalog every hour, add this to crontab (`crontab -e`):

```
0 * * * * docker exec -w /var/www/vhosts/localhost/html kerf3d_ols php spark drive:sync > /dev/null 2>&1
```

05 // 1-CLICK ZERO-DOWNTIME UPDATES

To update the site when new features or fixes are pushed to GitHub, simply run:

```
cd /var/www/kerf3d
chmod +x deploy.sh
./deploy.sh
```

06 // HANDOVER CHECKLIST FOR ADMINISTRATOR

ITEM	VALUE / DESCRIPTION
GitHub Repository	`https://github.com/redwan2003-bot/KERF_3D_store.git`
Live Domain	`kerf3d.store` (or designated domain)
bKash / Nagad Number	`01754550297`
Service Account JSON	`drive-service-account.json` (Delivered via secure private message)